

When Do Structural Primers Help LLMs Reason over Graphs?

Baseline Accuracy and Answer Leakage

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Abstract

A primer of structural facts, such as each node’s degree, is a natural aid for a language model answering questions about a graph given as text. We ask when such primers help, and why. We evaluate Qwen3-1.7B and Qwen3-4B, with and without thinking, on six GraphQA tasks over 40-node Erdős–Rényi graphs (84,000 generations). A solver that never sees the edges separates primers that state the answer from side information. On node-degree queries, a primer that states the answer adds about 16 points where accuracy without it is 25–75%, mostly for Qwen3-1.7B with thinking, and costs Qwen3-4B accuracy where that model is already accurate. The responses indicate why: the primer changes the procedure. Qwen3-4B mostly retrieves the stated degree instead of counting; it gains up to 27 points where its count is 33% accurate and loses up to 12 where it is 99% accurate. With thinking, Qwen3-1.7B checks its count against the stated degree and gains 16 to 20 points. On edge queries, primers change the rate of false “yes” answers: for Qwen3-1.7B, the combined primer lowers it by 19 points, and a preamble that only restates the node set raises it by 18. Primer length has no consistent effect.

1 Introduction

Fatemi et al. (2024) show that the accuracy of a large language model (LLM) on graph questions depends strongly on how the graph is written as text. A natural next step is to add derived structure to that text. We study *primers*: short preambles, placed before the graph encoding, that state each node’s degree, clustering coefficient or random-walk return probability, or the graph’s number of connected components. Graph neural networks take these statistics as input features (Dwivedi et al., 2022; Rampásek et al., 2022); here a pretrained LLM reads them as sentences.

Our research question is how primers change an LLM’s answers on graph tasks, and when they

help rather than harm. We divide it into three parts.

RQ1: how do primers that state the answer compare with primers that supply only side information, which does not determine it? **RQ2:** what decides whether a primer helps or harms a given model on a given task? **RQ3:** what does a primer change in the procedure the model follows?

Two controls make these questions answerable. A graph-blind solver reads the primer but never the edge list; where it is near-perfect, the primer states the answer (*answer leakage*), and a gain may reflect retrieval rather than reasoning (Geirhos et al., 2020). Because added text can itself degrade reasoning (Shi et al., 2023; Levy et al., 2024), a preamble that only restates the node set tests whether added text alone changes accuracy, and primers averaging 37 to 4,691 characters test whether effects follow length.

The results suggest that a primer which states the answer changes the procedure the model follows (Section 5). On node_degree such effects follow the accuracy without a primer (Section 4.1), one primer both helps and harms the same model (Section 4.2), and an answer-carrying primer helps only a model that can execute the procedure it implies (Section 4.3). Side information instead changes how often the model answers “yes” on edge queries and helps where its count fails (Section 4.4); bundling and length bring no consistent gain (Section 4.5).

2 Related Work

GraphQA (Fatemi et al., 2024), NLGraph (Wang et al., 2023), GPT4Graph (Guo et al., 2023) and GraphArena (Tang et al., 2025) evaluate LLMs on graph problems and vary the encoding, the prompt and the task. Sanford et al. (2024) show that four of our six tasks can be solved by a single transformer layer, yet prompted LLMs trail trained transformers on them. Prior work adds to the input an instruction to build the graph first (Wang et al., 2023), motif

Condition	States	Chars	Carries
none	no primer	–	–
filler	“Node i is simply present”	1,829	–
components	component count	37	–
clustering	clustering coefficient	1,629	–
rwse	return prob., 2 and 3 steps	2,949	–
degree	degree	891	deg., e.count
all	the three rows above	4,691	deg., e.count

Table 1: The seven conditions. *States*: what each primer gives per node (components: for the graph). *Chars*: mean length. *Carries*: tasks with variable gold on which the graph-blind solver scores at least 85% (Appendix Table 5): node_degree (deg.) and edge_count (e.count).

statistics (Das et al., 2024) or learned soft tokens (Perozzi et al., 2024). Huang et al. (2024) find that neighbourhood text in a prompt helps node classification through label-relevant phrases rather than through structure. We make the analogous separation for every task and condition, between text that determines the answer and text that does not. In the taxonomy of Rampásek et al. (2022), our primers state three local structural encodings, degree, triangle-based clustering and the random-walk return probability of Dwivedi et al. (2022), and one global structural encoding, the number of components.

The graph-blind solver is a partial-input baseline, as hypothesis-only models are for natural language inference (Gururangan et al., 2018; Poliak et al., 2018), applied here to an intervention rather than to a dataset; its failure does not rule out a shortcut (Feng et al., 2019).

3 Method

3.1 Primers and the graph-blind solver

A prompt consists of a primer, the graph in the incident encoding of Fatemi et al. (2024), and the question (Appendix A). Seven conditions share one renderer (Table 1). The five *content* primers state graph statistics, one sentence per node in id order, except components, which is a single sentence. filler writes “Node i is simply present within the graph G ” for each node: it restates the node set, which the encoding already lists, at a length 12% above that of clustering.

The graph-blind solver reads the rendered primer and the graph’s node and edge counts, except the count a task asks for, but never the edge list. It applies 16 exact rules where they exist and one heuristic or 8 fitted rules otherwise (fitted on sep-

arate graphs), so its accuracy is a lower bound on what the primer and the counts determine. We call a primer *answer-carrying* on a task when the solver scores at least 85% there, and *side information* otherwise; any threshold between 75% and 100% gives the same split (Table 1; Appendix Table 5).

3.2 Corpus, models and measurement

Graphs are Erdős–Rényi $G(n, p)$ graphs (Erdős and Rényi, 1959; Gilbert, 1959) with $n=40$ nodes at $p \in \{.10, .20, .35, .50\}$, 100 per density. The same graphs serve all six GraphQA tasks, and node_degree, connected_nodes and edge_existence query the same node. node_degree and edge_existence are extended to $p \in \{.65, .75, .85\}$. Four arms answer every prompt: Qwen3-1.7B and Qwen3-4B (Qwen Team, 2025b) with thinking mode off (*plain*: qwen3-1.7b, qwen3-4b) and on (*-think*), 84,000 generations. For qwen3-1.7b on node_degree we use three further runs: 300 new graphs per density, 1,600 graphs drawn with fresh seeds, and a grid that holds the mean degree at 8 or 16 while n ranges over $\{20, 40, 80, 160\}$. Decoding is greedy, with a budget of 8,192 new tokens (2,048 for the plain arms in the extension and the further runs).

Every task is scored by exact match; on connected_nodes, set-F1 moves a ninth to a sixth as much as exact match and reverses two of the five contrasts larger than 5 points. On edge_existence the share of queried pairs that are edges rises from 10% to 85% with density, so we also report the false-alarm rate (“yes” to a non-edge) and balanced accuracy. Each effect is a paired difference against none on the same graphs, in percentage points. A pair is dropped when either generation reaches the token budget (rates in Appendix Table 6). A *cell* is one (arm, task, density, content primer) combination; we keep the 393 cells that retain at least 50 pairs, and the 47 dropped are edge_count cells. node_count and cycle_check have a constant gold answer at $n=40$, so we exclude them from all summaries (Appendix B). Tests are exact McNemar (McNemar, 1947) with Benjamini–Hochberg correction (Benjamini and Hochberg, 1995) within each (arm, task), and intervals are 95% bootstrap intervals over graphs (Efron, 1979).

	qwen3-1.7b				qwen3-1.7b-think				qwen3-4b				qwen3-4b-think			
	exist.	deg.	conn.	e.count	exist.	deg.	conn.	e.count	exist.	deg.	conn.	e.count	exist.	deg.	conn.	e.count
none (acc.)	70.5	60.5	72.5	1.8	99.2	76.8	79.7	—	90.5	99.2	86.0	2.0	100.0	99.7	98.7	—
pairs	399	399	399	211	397	376	390	<i>13</i>	398	400	400	393	393	380	373	<i>34</i>
filler	-15.8	-0.5	-6.2	-0.4	-0.8	-3.5	+2.0	—	-7.0	-0.5	-12.5	+1.2	+0.0	+0.0	+0.8	—
components	-8.2	+1.8	+0.8	-1.4	-0.8	-1.0	+2.8	—	-2.8	+0.2	+9.2	-1.3	+0.0	-0.3	+0.5	—
clustering	+3.8	+3.0	+1.2	-0.4	-1.0	+1.0	+1.5	—	+4.0	-0.5	+1.0	-1.3	+0.0	-0.3	+0.0	—
rwse	+5.3	+4.0	+1.0	-0.8	-1.3	+2.5	-0.3	—	-10.0	-2.2	+0.0	-1.5	+0.0	-2.1	+0.0	—
degree	+4.0	+7.5	+3.5	+4.0	-2.3	+11.7	+7.4	—	-4.3	-6.5	-3.0	+27.7	+0.0	+0.0	+0.0	—
all	+11.5	+10.2	+2.0	+2.4	-2.0	+11.6	+6.1	—	-9.8	-8.0	-2.8	+3.8	+0.0	-0.8	-0.5	—

Table 2: Pooled effects against none (points), $p \leq .50$. Tasks: edge_existence (exist.), node_degree (deg.), connected_nodes (conn.), edge_count (e.count); the two constant-gold tasks are in Appendix Table 3. First row: accuracy (%) under none over terminated generations; *pairs*: fewest untruncated pairs. **Bold**: Benjamini-Hochberg-adjusted $q < .05$ (exact McNemar, six contrasts per column). Columns with *italic* pair counts (below 100) show no effects: their pairs are selected on termination.

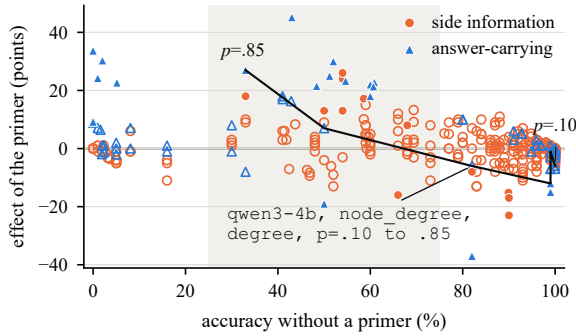


Figure 1: Effect of each content primer (every primer but filler) in each (arm, task, density) cell, against the cell’s accuracy without it. Triangles: answer-carrying; circles: side information; filled: $q < .05$, Benjamini-Hochberg over the cells of each (arm, task). The line joins the seven densities of degree for qwen3-4b on node_degree; shading marks baselines of 25–75%.

4 Results

4.1 Primer effects follow baseline accuracy

Figure 1 plots each cell’s effect against its accuracy without a primer, its *baseline*. Answer-carrying primers gain most at intermediate baselines: +16.0 and +16.7 points at baselines of 25–50% and 50–75%, against +7.8 below 25% and -5.8 between 75% and 90%. All 18 intermediate cells (-19 to +45) are node_degree cells, 10 of them from qwen3-1.7b-think. From 90% they cost 1.4 points on average, and every such cell of qwen3-4b loses accuracy (-1 to -15). Side information also peaks in the two middle bands, at a smaller scale: +2.3 and +3.9, against -1.8 below 25% and -0.8 from 90%. The pattern is not regression to the mean: binning each cell on half its graphs and measuring the effect on the other half reproduces every band to within about 3 points.

Outside edge_count, whose thinking-arm cells mostly truncate, thinking raises the median baseline from 67.0% to 94.9% on the 1.7B model and from 88.9% to 100.0% on the 4B model. It leaves 35 of the 1.7B model’s 90 cells between 25% and 75% and none of the 4B model’s, and no primer moves qwen3-4b-think by more than 2.1 points on node_degree, edge_existence or connected_nodes (Table 2).

Table 2 breaks the effects down by task: answer-carrying primers raise node_degree on both 1.7B arms (+7.5 to +11.7) and lower it on qwen3-4b (-6.5, -8.0), and the largest costs come from filler (-15.8, -12.5).

4.2 The sign follows the procedure the primer replaces

For qwen3-4b on node_degree, the degree primer states the queried degree. It changes accuracy by -6, -1, -7, -12 and -6 points at $p=.10$ to .65, and by +7 and +27 at .75 and .85 (Figure 1, line). The losses are not a cost of added text: at $p=.50$ the primer turns 13 correct answers into errors and corrects one, no generation truncates, and filler, which is longer, costs 2 points.

The responses show a change of procedure. We classify a node_degree response as *retrieval* when it answers without restating the queried node’s neighbour list, and as *enumeration* when it lists at least five neighbours one per line (Appendix Table 4). Without a primer, the baseline of qwen3-4b is the accuracy of its own count: 99–100% up to $p=.50$, then 82%, 50% and 33%. Under degree it retrieves in 44–74% of responses, at 73–98% accuracy: below the count up to $p=.50$ and above it from .65; enumeration falls from 85–100% of responses to at most 8% at $p=.35-.75$. From $p=.50$

most other responses (68–92%) open with an answer, usually not the stated degree, and then recount, at 79%, 45%, 28% and 16% accuracy; at .65 they outweigh retrieval’s gain. Retrieval is more accurate for nodes 0–9 (96%, against 83%; +12 points within density, significant at .001 by permutation), whose lines open the primer but whose ids are also the only single-digit ones; from node 10 on, accuracy does not fall with position. Under all, which states the degree among two other statistics, retrieval falls to 15–51% of responses at 49–96% accuracy, the other responses are 32%, 12% and 5% accurate from $p=.65$, and accuracy drops at every density, by up to 37 points.

The thinking arm uses the same primer differently: under degree, qwen3-1.7b-think retrieves in at most 2% of responses and, from $p=.35$ on, enumerates the neighbours in 76–98%. It reports a discrepancy between its count and the primer in 23% of responses up to $p=.50$ and in 49% beyond (without a primer, 0.3% and 4% report any discrepancy), and it is correct in 61% and 69% of those. The primer adds 19.9 points [16.3, 23.6] over the seven densities, with little change at high baselines (+5.2 and -1.0 at $p=.10$ and $.20$). It also makes 8% of generations reach the budget (1% without a primer; 96% of them report a discrepancy); counting these as errors, the gain is 16.0 [12.0, 19.7]. The plain qwen3-1.7b rarely retrieves (at most 4% of responses), yet gains 10 and 18 points at $p=.20$ and $.35$, where its count is 80% and 41% accurate; from $p=.50$, where its baseline is 5–30%, it stays within 2 points.

4.3 An answer-carrying primer helps only through a procedure the model can execute

On edge_count, degree states every degree, so the answer is half their sum. Without a primer qwen3-4b already sets out to sum the degrees (96–100% of responses) but is exact on only 2% of graphs. Given the degrees it is exact on 30% of graphs (+23 to +34 points at each density; 114 items corrected, 5 broken). qwen3-1.7b takes up the same procedure (from 8–64% of responses to 99–100%) but rarely completes the forty-term sum correctly: 4.7% of its answers are exact, a gain of 4.0 points that does not survive correction, although its mean absolute error falls from 136 to 40 edges.

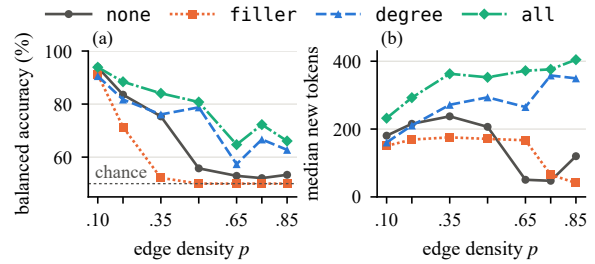


Figure 2: qwen3-1.7b on edge_existence under four of the seven conditions: (a) balanced accuracy and (b) median new tokens by density. Without a primer the model answers “yes” to almost every pair from $p=.65$; under filler, from $p=.35$.

4.4 Side information shifts false alarms and helps where counting fails

On edge_existence the plain arms answer “yes” to 97–100% of true edges under every condition, and 94–99% of their errors are false alarms, so primers change accuracy almost entirely through the false-alarm rate. Pooled over the seven densities, all lowers the false-alarm rate of qwen3-1.7b from 51% to 33% (-18.6 points [$-24.0, -13.1$]), and degree, rwse and clustering lower it by 6 to 8 points. By contrast, filler raises it to 69% (+18.3 [13.9, 22.7]) and components to 61%. For qwen3-4b, clustering is the only primer that lowers the false-alarm rate (16% to 8%) and the only one that raises accuracy on the task (+4.6 [2.3, 6.9]); the other five conditions raise false alarms by 7 to 14 points.

At high density qwen3-1.7b defaults to “yes” (Figure 2). Without a primer it answers “yes” to 98–99% of pairs from $p=.65$, in short responses, at a balanced accuracy of 52–53%. The two primers that state degrees keep the responses long and restore part of the balanced accuracy (degree +9.5 [4.5, 15.0], all +14.9 [8.7, 21.4]). Under filler it defaults to “yes” from $p=.35$ (97% of pairs there, against 65% without a primer).

Side information also helps where counting fails: on qwen3-4b/node_degree above $p=.50$, clustering adds 11.3 points [6.7, 16.3] (46 items corrected, 12 broken) and rwse 7.7, whereas filler adds 2.0 (not significant). These primers state no degree; under clustering the model keeps its procedure (69% of responses enumerate, against 74% without a primer), and the printed coefficients vary little within a graph (mean standard deviation at most 0.02), so the gain does not rest on node-specific values. clustering also helps

qwen3-1.7b: +3.0 in the main sweep, replicated at +4.0, +4.2 and +6.2 points in the three further runs (each significant at .005).

Some gains need no graph-specific content: components raises qwen3-4b on connected_nodes by 9 to 11 points at $p \geq .20$, where every graph is connected and the primer is one fixed 37-character sentence, and by 7 at $p=.10$.

4.5 No consistent gain from bundling or length

all states every value that degree, clustering and rwse state, yet its effect exceeds the mean of theirs by only 0.2 points ($[-1.1, 1.6]$, bootstrap over 54 (arm, task, density) combinations), and on qwen3-4b/node_degree above $p=.50$, where each part gains on average (Sections 4.2 and 4.4), it costs 21.3 points.

Primer length does not order the effects consistently. Regressed on the lengths of the five content primers, the effects in each column of Table 2 give slopes of -2.0 to $+3.4$ points per 1,000 characters. The slope is positive in 6 of 7 fits for the 1.7B model and none of 7 for the 4B model (2 and 0 without all, the longest primer). filler lies 18.3 and 13.8 points below the fitted line where it harms most. On node_count, which the summaries exclude, qwen3-1.7b answers 39 instead of 40 on 68.5% of items without a primer, 1.2% under clustering and 66.5% under the longer rwse (Appendix Table 3).

5 Discussion and Conclusion

For primers that state the answer, the results suggest one account: the primer changes the procedure the model follows, and its effect depends on how accurate the new procedure is relative to the count it replaces. Without a primer the model must count, so its baseline measures the accuracy of the count a primer can displace. What replaces the count depends on the model: retrieval and a less accurate recount for qwen3-4b, a check against the stated degree for the thinking 1.7B arm. Side information states no answer to retrieve, so on node_degree it can act only on the count.

Two results bear on prior work. A single transformer layer can solve node_degree and edge_count (Sanford et al., 2024), yet given the answer qwen3-4b is right on only 87% of node_degree items at $p=.50$, and both plain arms

mostly fail the forty-term sum that the stated degrees leave on edge_count (Section 4.3), consistent with the compositional failures Dziri et al. (2023) describe. For a prompted model the difficulty appears to lie in the procedure, not the information. The text controls suggest that added text matters through what it states: restating the node set raises false “yes” answers, consistent with Shi et al. (2023), while length, which varies only together with content, shows no consistent cost of the kind Levy et al. (2024) report.

In answer to our questions: **RQ1**: at intermediate baselines, a stated answer adds about 16 points on node_degree, and side information 2 to 4, mostly from degree and all on edge_existence and connected_nodes. **RQ2**: a stated answer helps where the procedure it induces is more accurate than the model’s own count; side information helps where that count is partly wrong. **RQ3**: a primer changes the procedure, to retrieval or a check on node_degree, and on edge queries it shifts the model’s bias for answering “yes”, leaving hits unchanged. Evaluations of graph prompts should report each cell’s baseline, a graph-blind solver’s score and, for yes/no tasks, the false-alarm rate.

Limitations

We evaluate two model sizes from one family on Erdős–Rényi graphs, with one encoding, template, node order and primer position, although position and order affect accuracy (Liu et al., 2024; Ge et al., 2025). Decoding is greedy, whereas the Qwen3 model card (Qwen Team, 2025a) recommends sampling and 32,768 output tokens, so the thinking arms’ truncation on edge_count (81% and 62%) may be specific to this protocol. Regenerating 1,200 identical qwen3-1.7b prompts changes 5% of outcomes, so part of each cell’s discordance is run-to-run variation. Where side information acts (baselines of 25–75%), a cell of 100 graphs detects paired effects of about 14 points, so those results rest on averages over cells. Procedures are classified by surface patterns, and a model may follow a procedure it does not state (Turpin et al., 2023).

Ethics Statement

We evaluate open-weight models on synthetic graphs, without human subjects, personal data or annotation, at a GPU cost of about 217,000 generations (97,200 analysed here).

References

- Yoav Benjamini and Yosef Hochberg. 1995. [Controlling the false discovery rate: A practical and powerful approach to multiple testing](#). *Journal of the Royal Statistical Society: Series B (Methodological)*, 57(1):289–300.
- Debarati Das, Ishaan Gupta, Jaideep Srivastava, and Dongyeop Kang. 2024. [Which modality should I use – text, motif, or image? : Understanding graphs with large language models](#). In *Findings of the Association for Computational Linguistics: NAACL 2024*, pages 503–519.
- Vijay Prakash Dwivedi, Anh Tuan Luu, Thomas Laurent, Yoshua Bengio, and Xavier Bresson. 2022. [Graph neural networks with learnable structural and positional representations](#). In *International Conference on Learning Representations*.
- Nouha Dziri, Ximing Lu, Melanie Sclar, Xiang Lorraine Li, Liwei Jiang, Bill Yuchen Lin, Sean Welleck, Peter West, Chandra Bhagavatula, Ronan Le Bras, Jena D. Hwang, Soumya Sanyal, Xiang Ren, Allyson Ettinger, Zaid Harchaoui, and Yejin Choi. 2023. [Faith and fate: Limits of transformers on compositionality](#). In *Advances in Neural Information Processing Systems*, volume 36, pages 70293–70332.
- Bradley Efron. 1979. [Bootstrap methods: Another look at the jackknife](#). *The Annals of Statistics*, 7(1):1–26.
- Paul Erdős and Alfréd Rényi. 1959. [On random graphs I](#). *Publicationes Mathematicae Debrecen*, 6(3–4):290–297.
- Bahare Fatemi, Jonathan Halcrow, and Bryan Perozzi. 2024. [Talk like a graph: Encoding graphs for large language models](#). In *International Conference on Learning Representations*, pages 43909–43934.
- Shi Feng, Eric Wallace, and Jordan Boyd-Graber. 2019. [Misleading failures of partial-input baselines](#). In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 5533–5538.
- Yuyao Ge, Shenghua Liu, Baolong Bi, Yiwei Wang, Lingrui Mei, Wenjie Feng, Lizhe Chen, and Xueqi Cheng. 2025. [Can graph descriptive order affect solving graph problems with LLMs?](#) In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 6404–6420.
- Robert Geirhos, Jörn-Henrik Jacobsen, Claudio Michaelis, Richard Zemel, Wieland Brendel, Matthias Bethge, and Felix A. Wichmann. 2020. [Shortcut learning in deep neural networks](#). *Nature Machine Intelligence*, 2(11):665–673.
- E. N. Gilbert. 1959. [Random graphs](#). *The Annals of Mathematical Statistics*, 30(4):1141–1144.
- Yiayan Guo, Lun Du, Hengyu Liu, Mengyu Zhou, Xinyi He, and Shi Han. 2023. [GPT4Graph: Can large language models understand graph structured data? An empirical evaluation and benchmarking](#). *arXiv preprint arXiv:2305.15066*.
- Suchin Gururangan, Swabha Swayamdipta, Omer Levy, Roy Schwartz, Samuel R. Bowman, and Noah A. Smith. 2018. [Annotation artifacts in natural language inference data](#). In *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 2 (Short Papers)*, pages 107–112.
- Jin Huang, Xingjian Zhang, Qiaozhu Mei, and Jiaqi Ma. 2024. [Can LLMs effectively leverage graph structural information through prompts, and why?](#) *Transactions on Machine Learning Research*.
- Mosh Levy, Alon Jacoby, and Yoav Goldberg. 2024. [Same task, more tokens: the impact of input length on the reasoning performance of large language models](#). In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 15339–15353.
- Nelson F. Liu, Kevin Lin, John Hewitt, Ashwin Paranjape, Michele Bevilacqua, Fabio Petroni, and Percy Liang. 2024. [Lost in the middle: How language models use long contexts](#). *Transactions of the Association for Computational Linguistics*, 12:157–173.
- Quinn McNemar. 1947. [Note on the sampling error of the difference between correlated proportions or percentages](#). *Psychometrika*, 12(2):153–157.
- Bryan Perozzi, Bahare Fatemi, Dustin Zelle, Anton Tsitsulin, Mehran Kazemi, Rami Al-Rfou, and Jonathan Halcrow. 2024. [Let your graph do the talking: Encoding structured data for LLMs](#). *arXiv preprint arXiv:2402.05862*.
- Adam Poliak, Jason Naradowsky, Aparajita Haldar, Rachel Rudinger, and Benjamin Van Durme. 2018. [Hypothesis only baselines in natural language inference](#). In *Proceedings of the Seventh Joint Conference on Lexical and Computational Semantics*, pages 180–191.
- Qwen Team. 2025a. Qwen3-4B model card. <https://huggingface.co/Qwen/Qwen3-4B>.
- Qwen Team. 2025b. [Qwen3 technical report](#). *arXiv preprint arXiv:2505.09388*.
- Ladislav Rampášek, Michael Galkin, Vijay Prakash Dwivedi, Anh Tuan Luu, Guy Wolf, and Dominique Beaini. 2022. [Recipe for a general, powerful, scalable graph transformer](#). In *Advances in Neural Information Processing Systems*, volume 35, pages 14501–14515.
- Clayton Sanford, Bahare Fatemi, Ethan Hall, Anton Tsitsulin, Mehran Kazemi, Jonathan Halcrow, Bryan Perozzi, and Vahab Mirrokni. 2024. [Understanding](#)

transformer reasoning capabilities via graph algorithms. In *Advances in Neural Information Processing Systems*, volume 37, pages 78320–78370.

Freda Shi, Xinyun Chen, Kanishka Misra, Nathan Scales, David Dohan, Ed H. Chi, Nathanael Schärli, and Denny Zhou. 2023. [Large language models can be easily distracted by irrelevant context](#). In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pages 31210–31227.

Jianheng Tang, Qifan Zhang, Yuhua Li, Nuo Chen, and Jia Li. 2025. [GraphArena: Evaluating and exploring large language models on graph computation](#). In *International Conference on Learning Representations*, pages 48118–48145.

Miles Turpin, Julian Michael, Ethan Perez, and Samuel Bowman. 2023. [Language models don’t always say what they think: Unfaithful explanations in chain-of-thought prompting](#). In *Advances in Neural Information Processing Systems*, volume 36, pages 74952–74965.

Heng Wang, Shangbin Feng, Tianxing He, Zhaoxuan Tan, Xiaochuang Han, and Yulia Tsvetkov. 2023. [Can language models solve graph problems in natural language?](#) In *Advances in Neural Information Processing Systems*, volume 36, pages 30840–30861.

A Example prompt

A clustering-condition node_degree prompt at $p=.10$ (elisions marked ...; line breaks added for width). Other conditions replace the first paragraph; none omits it.

Node 0 has clustering coefficient 0.20. Node 1 has clustering coefficient 0.00. ... Node 39 has clustering coefficient 0.33.

G describes a graph among nodes 0, 1, 2, ..., 38, and 39.
In this graph:
Node 0 is connected to nodes 2, 6, 8, 12, 35.
Node 1 is connected to nodes 5, 20.
...
Node 39 is connected to nodes 15, 16, 18, 20.
Q: What is the degree of node 16?
A:

B Constant-gold tasks

Arm	none	filler	comp.	clust.	rwse	degree	all
node_count							
1.7B	31.5	+39.8	+21.2	+67.2	+2.0	+6.8	+47.0
1.7B-think	98.6	+1.4	+0.9	+1.4	+1.4	+0.8	+1.4
4B	90.2	+9.2	+9.0	+8.8	+9.2	+8.0	+9.8
4B-think	99.5	+0.5	+0.0	+0.5	+0.5	+0.5	+0.5
cycle_check							
1.7B	99.7	+0.3	−0.6	+0.3	−7.5	−12.6	−1.1
1.7B-think	100.0	+0.0	+0.0	+0.0	+0.0	−0.7	+0.0
4B	99.7	−0.3	−1.3	+0.3	+0.3	−1.5	−0.5
4B-think	100.0	+0.0	+0.0	+0.0	+0.0	−0.8	+0.0

Table 3: The two tasks with constant gold at $n=40$, pooled over $p \leq .50$: accuracy under none (%) and paired effects against it (points). **Bold**: as in Table 2.

C Procedure on node_degree

p	.10	.20	.35	.50	.65	.75	.85
qwen3-4b							
acc., none	100	99	99	99	82	50	33
enumerates	1	34	95	100	94	85	44
acc., degree	94	98	92	87	76	57	60
retrieves	44	56	74	62	71	64	69
acc. of retrievals	89	98	91	92	89	73	80
acc. of the rest	98	98	96	79	45	28	16
enumerates	2	16	6	8	2	1	0
acc., all	93	96	92	84	45	31	25
retrieves	15	24	18	16	47	51	34
acc. of retrievals	93	96	78	88	60	49	65
acc. of the rest	93	96	95	83	32	12	5
qwen3-1.7b-think							
acc., none	93	95	60	60	52	47	42
acc., degree	98	94	84	82	78	76	87
enumerates	2	37	76	91	90	98	96
reports a discrepancy	6	10	40	39	59	47	40
qwen3-1.7b							
acc., none	91	80	41	30	8	16	5
acc., degree	94	90	59	29	8	15	7
retrieves	4	2	0	0	0	0	4

Table 4: Procedure on node_degree by density (%; terminated generations). *Retrieves*: the response answers without restating the queried node’s neighbour list. *Enumerates*: it lists at least five neighbours one per line. *The rest*: responses that do not retrieve. *Reports a discrepancy*: it states that its count and the primer disagree (a check for one does not count). Without a primer there is no stated degree to retrieve, so the none rows give the accuracy of the model’s own count.

D Solver scores and truncation

Task	none	filler	comp.	clust.	rwse	degree	all
node_count	100	100	100	100	100	100	100
edge_count	3.0	3.0	3.0	10.4	3.0	100	100
node_degree	14.4	14.4	14.4	14.4	21.0	100	100
connected_nodes	0.0	0.0	0.0	0.0	0.1	0.1	0.8
edge_existence	73.5	73.5	73.5	72.7	55.4	74.6	74.1
cycle_check	100	100	100	100	100	100	100

Table 5: Accuracy (%) of the graph-blind solver at $n=40$, averaged over seven densities (300 test graphs per density; fitted rules use separate graphs). It reads the rendered primer and the node and edge counts, except the count a task asks for, never the edge list. **Bold**: answer-carrying (at least 85%) on a task whose gold varies. On node_count and cycle_check every score is 100 because the gold answer is constant at $n=40$. On edge_existence, rwse scores below none: from $p=.65$ its fitted density rule is less accurate than the majority answer, on the fitting graphs as well.

Task	1.7B	1.7B-think	4B	4B-think
edge_count	22.4%	81.0%	0.6%	62.2%
cycle_check	6.1%	6.5%	1.0%	1.9%
node_degree	0.0%	1.1%	0.0%	1.6%
connected_nodes	0.0%	0.8%	0.0%	3.5%
edge_existence	0.0%	0.2%	0.1%	0.8%
node_count	0.0%	2.5%	0.0%	0.1%

Table 6: Share of main-sweep generations ($p \leq .50$) that reach the 8,192-token budget, by task and arm, pooled over condition and density. **Bold:** above 50%. In the high-density extension no (task, arm) exceeds 2.5%. Under degree, qwen3-1.7b-think reaches the budget on 7.6% of its node_degree generations over the seven densities, against 1.1% without a primer. Over all 84,000 generations, all but one of the 78,538 that terminate yield an answer.